

Realistic Rendering Tool for Pseudo-Structural Coloring with Multi-Color Extrusion of FFF 3D Printing

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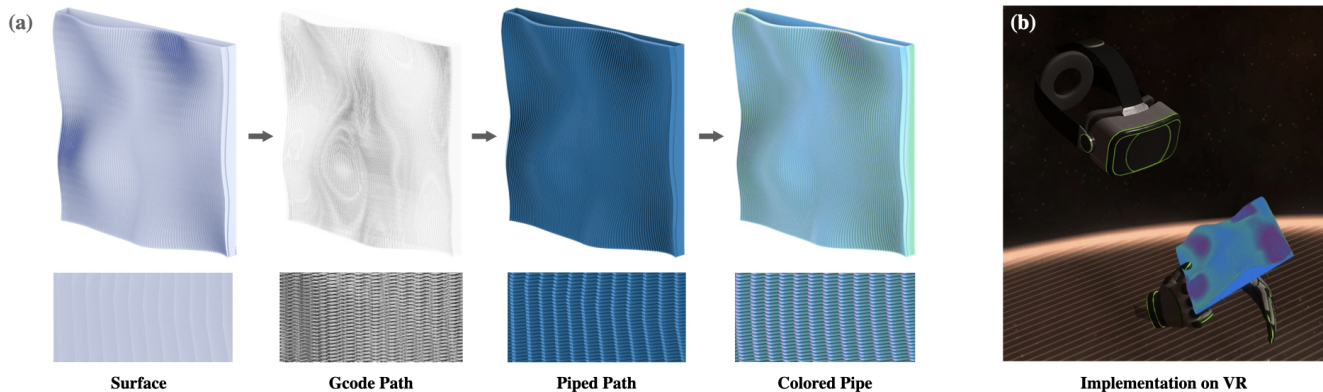


Figure 1: (a)The gcode generated from the surface is piped by importing it into our original tool. The tool then colors according to the normal direction. (b)The generated model allows the viewer to experience the transition of colors when moving it by viewing on VR.

ABSTRACT

In this study, we developed a flow to simulate anisotropy in multi-color printing of a FFF 3D printer. There are previous methods for simulating layered marks from gcode, which is a 3D printer control statement, and multi-color printing by switching filament colors and their simulators. However, there was no simulator for the case where color anisotropy exists in a single path, which is the target of this study. Since the coloring of this method is based on a completely different principle from existing methods, it is difficult to predict the results. Our simulator tool was very useful in rendering multi-color 3D printed models. Furthermore, a more sophisticated simulation of the placement of objects in real space was performed based on the highly realistic images created by physical-based rendering. The simulation also worked in VR space and showed an optimal preview at the architectural scale.

CCS CONCEPTS

• Human-centered computing → Accessibility theory, concepts and paradigms.

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KEYWORDS

3D printing, gcode Simulation, multi-color, digital fabrication

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1 INTRODUCTION

FFF (Fused Filament Fabrication) 3D printing is becoming widely used in the design and art fields. The layered textures that appear through 3D printing are used in the design. In order to control layered textures, the software is also being developed to generate coordinate control files to run the 3D printer, called gcode. [Jasche and Ludwig 2020] introduces a tool that uses gcode to check in advance what textures and shapes will appear.

On the other hand, the advent of multi-color filaments has enabled 3D printing with coloring according to the direction in which the surface faces. The visible color varies greatly depending on the shape of the model. Sekisai has also used this technology to create an impressive wall work called Transpective [sekisai.com 2021]. However, this work does not provide predictions for multi-colored models from gcode. In terms of color gcode prediction, [Takahashi et al. 2020] introduces the gcode viewer of 3D printing while switching colors.

In this study, we propose the method to generate the colors and shapes with layered textures like 3D printing from gcode. This allows the appearance and color transition to be checked as design and the art production to proceed through multi-color 3D printing. The generated 3D printed-like mesh model allows the physical-based rendering to produce realistic images. The mesh brought into the VR space allows the user to freely move their body and see the color transition(Figure 1(b)).

2 OUR MODEL

Figure 2 shows the actual behavior of the plastic when the three colored filaments are extruded simultaneously. Different colors are visible because of the different directions of the surfaces in each location. We have developed a tool to simulate the colors appearing at each position.

Our tool has four steps shown in Figure 1(a). First, we prepare the gcode data generated from the surface model. We load the gcode into our original tool on Houdini. This tool generates lines between points based on the gcode extrusion value. The lines are replaced with flattened pipes to reproduce the 3D printed layered shape. Three colors are painted according to the normal direction of these pipe meshes to provide the multi-color effect of 3D printing.

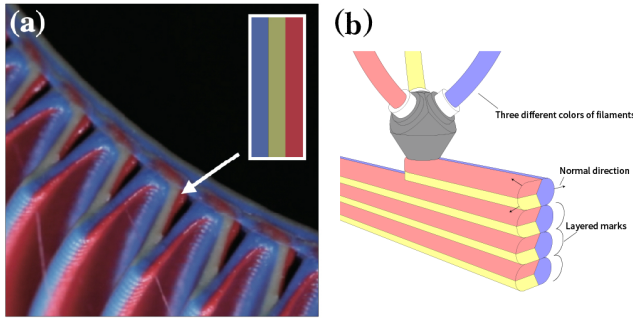


Figure 2: (a)Top view of the model with three different colors depending on the normal direction of the surface, (b) Diagram with color change on top and bottom of the layer marks.

3 IMPLEMENTATION

3.1 Physical Based Rendering(PBR)

We present a realistic image by PhysicalBasedRendering (PBR) as an implementation of the generated data. Realistic and detailed renderings help check the colors created by multi-color 3D printing(Figure 3). In addition, applying rotations and other animations on the CG software allows the user to check color transitions when the angle of view changes.

3.2 Preview In Virtual Reality

When rendering the anisotropy of coloration on an architectural scale, it is important to give the viewer the virtual experience with highly physical and spatial feelings. For example, in desktop applications using existing displays, the design object is rotated and

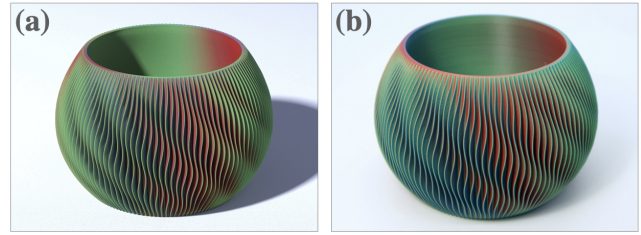


Figure 3: (a) The rendered model image generated by our tool. (b) The actual 3D printed model image used the same gcode.

examined from a bird's eye view. However, in the case of an actual installed architectural-scale modeling object, the subject of movement is the viewer themselves. In other words, previewing with existing software is only an indirect spatial representation, leaving no room for physicality to intervene, and is not suitable for previewing coloring on an architectural scale. Therefore, in this study, we made the simulation with spatiality and physicality that possible by importing the mesh generated from gcode into NeosVR, a social VR platform.

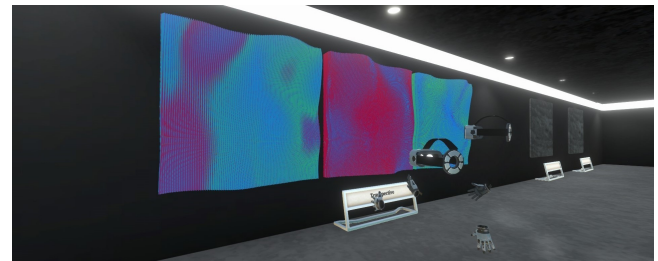


Figure 4: A preview on NeosVR.

4 CONCLUSION AND FUTUREWORKS

In this study, we introduced the tool that allowed coloring according to the normal direction of the mesh by loading gcode. We also introduced implementations using our tool. Once gcode was created, it was possible to simulate the colors that would emerge from a multi-color 3D print and to check the appearance without actual 3D printing. It was also possible to check how the colors transitioned when the model or person moved. Future work includes feedback from the simulation results to gcode design. For example, we expect to develop a modeling tool that can be used to fine-tune the shapes based on the simulated visuals.

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